

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 1 – Data Interpretation

|                  |                                                                                                                                   |               |                                         |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------|
| Course Code:     | DSA03                                                                                                                             | Course Title: | Natural Language Processing             |
| CO Assessed:     | CO4 – Precision                                                                                                                   | Assessment:   | Assessment Tool 1 – Data Interpretation |
| Weightage:       | 40% of CIA                                                                                                                        |               |                                         |
| CO5 Statement:   | Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4) |               |                                         |
| Student Name:    | N.Kishan Kumar                                                                                                                    | Reg. No.:     | 192424404                               |
| Section / Batch: | B.Tech AI & DS                                                                                                                    | Date:         | 08-08-2026                              |

### Code of Conduct

I \_\_N.Kishan kumar-1924244404\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: \_\_\_\_\_

### Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

| Section / Topic                                                                | Focus                                                                                                                                                         | Question No. |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Representing Meaning & Meaning Structure of Language                           | Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems. | Q1           |
| First-Order Predicate Calculus & Representing Linguistically Relevant Concepts | Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.           | Q2           |
| Lexemes and Their Senses & Word Sense Disambiguation                           | Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.                 | Q3           |
| Syntax-Driven Semantic Analysis                                                | Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.           | Q4           |

# ASSESSMENT TOOL 1 – DATA INTERPRETATION

## Q1. Semantic Representation in Customer Support Chatbots

### 1. Analyze the semantic representations and identify the action-object relationships.

Semantic representation converts a user's natural language query into a structured meaning representation containing an **action** and an **object/entity**. The given chatbot representations are:

| User Query                                    | Semantic Representation          | Action     | Object       |
|-----------------------------------------------|----------------------------------|------------|--------------|
| Activate international roaming for my number. | ACTIVATE(Roaming, Customer)      | ACTIVATE   | Roaming      |
| Deactivate caller tune service.               | DEACTIVATE(CallerTune, Customer) | DEACTIVATE | Caller Tune  |
| Check my data balance.                        | QUERY(DataBalance, Customer)     | QUERY      | Data Balance |
| Enable 5G service.                            | ACTIVATE(5GService, Customer)    | ACTIVATE   | 5G Service   |

The representations correctly capture the main intention of the customer. **ACTIVATE** indicates enabling a service, **DEACTIVATE** indicates disabling a service, and **QUERY** represents a request for information. The second argument, Customer, identifies the user associated with the requested operation. The semantic representations provided in the question are shown on page 2 of the assessment.

### 2. Determine which query contains semantic interpretation errors.

The chatbot log is:

| Query ID | Actual User Intent     | Predicted Intent     | Result           |
|----------|------------------------|----------------------|------------------|
| Q1       | Activate Roaming       | Activate Roaming     | Correct          |
| Q2       | Deactivate Caller Tune | Activate Caller Tune | <b>Incorrect</b> |
| Q3       | Check Data Balance     | Query Data Balance   | Correct          |
| Q4       | Enable 5G Service      | Activate 5G Service  | Correct          |

### Q2 contains the semantic interpretation error.

The actual intent is **DEACTIVATE Caller Tune**, whereas the predicted intent is **ACTIVATE Caller Tune**. The object, Caller Tune, is correctly identified, but the action is incorrectly classified.

Therefore:

**Actual:** DEACTIVATE(CallerTune, Customer)

**Predicted:** ACTIVATE(CallerTune, Customer)

This is a significant semantic error because activation and deactivation represent opposite operations. The chatbot log confirms this mismatch.

### 3. Evaluate how meaning representation affects chatbot decision accuracy.

Meaning representation directly affects the decision made by the chatbot. A correct representation allows the system to identify:

- What action the customer wants.
- Which service or entity is involved.
- Which customer the operation applies to.
- What backend operation should be performed.

For Q1, Q3 and Q4, the semantic representation correctly identifies the user's intention, allowing the chatbot to select the appropriate operation.

However, Q2 demonstrates that an incorrect action representation can cause an incorrect service operation. If the system interprets "Deactivate caller tune" as "Activate caller tune," the chatbot may perform the opposite action.

Thus, semantic analysis acts as an important bridge between natural language input and system action.

#### 4. Recommendations to improve semantic understanding

The following improvements can increase chatbot accuracy:

1. **Intent classification:** Use supervised machine-learning or transformer-based models to classify user intentions.
2. **Action-object extraction:** Explicitly identify actions such as Activate, Deactivate and Query and associate them with the correct service.
3. **Context awareness:** Maintain previous conversation context to resolve incomplete or ambiguous requests.
4. **Synonym handling:** Recognize expressions such as *enable/activate* and *disable/deactivate*.
5. **Confidence checking:** If the predicted intent has low confidence, ask the customer for confirmation.
6. **Continuous training:** Use incorrectly classified customer queries as training examples for future improvement.

#### Conclusion

The analysis shows that **Q2 is the only incorrect semantic interpretation** among the four queries. The problem is caused by confusing the action **DEACTIVATE** with **ACTIVATE** while correctly identifying the Caller Tune object. Therefore, accurate semantic representation is essential for reliable chatbot decision-making and preventing incorrect telecom service operations. The assessment specifically asks for analysis of action-object relationships, interpretation errors, chatbot accuracy, and improvements.

## Q2. First-Order Predicate Calculus for Smart Manufacturing

The manufacturing plant provides the following machine statuses:

| Machine | Status      |
|---------|-------------|
| M1      | Active      |
| M2      | Active      |
| M3      | Maintenance |
| M4      | Active      |

The given predicate rules are:

- **Active(x) → Producing(x)**
- **Produces(x,y) ∧ Active(x) → Available(y)**
- **Maintenance(x) → ¬Producing(x)**

These rules and machine states are given in the assessment scenario.

#### 1. Represent the production data using First-Order Predicate Calculus.

The production data can be represented as:

**Active(M1)**

**Active(M2)**

**Maintenance(M3)**

**Active(M4)**

Using the rule:

**Active(x)  $\rightarrow$  Producing(x)**

we derive:

**Producing(M1)**

**Producing(M2)**

**Producing(M4)**

Using:

**Maintenance(x)  $\rightarrow$   $\neg$ Producing(x)**

we derive:

**$\neg$ Producing(M3)**

Therefore, the current logical state is:

| Machine | Predicate       | Inference            |
|---------|-----------------|----------------------|
| M1      | Active(M1)      | Producing(M1)        |
| M2      | Active(M2)      | Producing(M2)        |
| M3      | Maintenance(M3) | $\neg$ Producing(M3) |
| M4      | Active(M4)      | Producing(M4)        |

## 2. Analyze which products are currently available.

The rule for product availability is:

**Produces(x,y)  $\wedge$  Active(x)  $\rightarrow$  Available(y)**

This means a product y is available only when:

1. Machine x produces y, and
2. Machine x is active.

However, the given data does not specify which machine produces which product. No facts such as:

**Produces(M1,Gear)**

are provided.

Therefore, the available products **cannot be determined completely from the given data.**

For example, if the additional fact were:

**Produces(M1,Gear)**

then:

**Produces(M1,Gear)  $\wedge$  Active(M1)**

would allow us to infer:

**Available(Gear)**

This demonstrates how predicate logic combines production relationships and machine status to determine availability.

## 3. Infer whether Gear production is affected by maintenance activities.

The given information states:

**Maintenance(M3)  $\rightarrow$   $\neg$ Producing(M3)**

Therefore:

**$\neg$ Producing(M3)**

can be concluded.

However, the question does not provide a fact connecting **M3** to **Gear**, such as:

**Produces(M3,Gear)**

Therefore, it is not logically valid to conclude that Gear production is affected by M3's maintenance.

If the missing fact were:

**Produces(M3,Gear)**

then we could conclude that M3 is not producing Gear during maintenance, and therefore Gear production through M3 is affected.

Hence, based strictly on the provided information:

**Conclusion: Gear production impact cannot be definitively established because the machine-to-product relationship is not provided.**

This careful conclusion is important because predicate logic should derive conclusions only from available facts and valid rules.

#### **4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.**

First-Order Predicate Calculus is useful in smart manufacturing because it provides a formal representation of machine states and relationships.

Its advantages include:

- Represents machine conditions using precise predicates.
- Supports logical inference from existing production data.
- Detects conditions such as active and maintenance states.
- Helps determine production and product availability.
- Provides explainable decision-making because conclusions follow explicit rules.

For example:

**Active(M1) → Producing(M1)**

clearly explains why M1 is considered a producing machine.

However, predicate logic also depends strongly on the completeness and correctness of input facts. In this scenario, product availability cannot be determined because **Produces(x,y)** relationships are missing.

#### **Conclusion**

Predicate calculus provides a structured and explainable method for manufacturing decision support. From the given data, M1, M2 and M4 are producing, while M3 is not producing because it is under maintenance. However, the availability of specific products such as Gear cannot be determined without machine-product relationships. Therefore, predicate logic is effective, but its conclusions depend on complete and accurate production data.

## **Q3. Word Sense Disambiguation in E-Commerce Search Engines**

Word Sense Disambiguation (WSD) is the process of identifying the correct meaning of an ambiguous word using its surrounding context.

The assessment provides four ambiguous queries and their possible senses.

#### **1. Analyze the contextual information and determine the correct word sense.**

| Query             | Possible Senses          | Clicked Result  | Correct Sense           |
|-------------------|--------------------------|-----------------|-------------------------|
| Apple accessories | Fruit / Technology Brand | iPhone Charger  | <b>Technology Brand</b> |
| Mouse wireless    | Animal / Computer Device | Bluetooth Mouse | <b>Computer Device</b>  |

|               |                               |                               |                             |
|---------------|-------------------------------|-------------------------------|-----------------------------|
| Java tutorial | Island / Programming Language | Coding Lessons                | <b>Programming Language</b> |
| Python course | Snake / Programming Language  | Software Development Training | <b>Programming Language</b> |

The clicked search results provide strong contextual evidence for determining the intended meaning.

For "Apple accessories," the result is an iPhone charger, indicating that Apple refers to the technology brand rather than fruit.

For "Mouse wireless," the result is a Bluetooth mouse, so Mouse refers to a computer device rather than an animal.

For "Java tutorial," the result is coding lessons, indicating that Java refers to the programming language rather than the island.

For "Python course," software development training indicates that Python refers to the programming language rather than the snake.

## 2. Identify the semantic cues used for disambiguation.

The major semantic cues are:

1. **Query words:** Words such as *accessories*, *wireless*, *tutorial*, and *course* provide contextual information.
2. **Clicked results:** User interaction gives direct evidence about the intended meaning.
3. **Domain context:** E-commerce and software-related contexts narrow the possible senses.
4. **Co-occurring terms:** Words appearing with the ambiguous term help determine its meaning.
5. **User behaviour:** Click history provides useful evidence for personalization and sense selection.

For example:

**Python + course + software development → Programming Language**

rather than:

**Python + animal-related context → Snake**

## 3. Evaluate the impact of incorrect sense selection.

Incorrect WSD can significantly reduce search-engine performance.

For example, if "**Java tutorial**" is interpreted as an island instead of a programming language, the search engine may return travel information rather than programming tutorials.

Similarly, if "**Apple accessories**" is interpreted as fruit, the system may return kitchen or agricultural products instead of iPhone accessories.

Incorrect sense selection can result in:

- Irrelevant search results.
- Lower click-through rates.
- Poor recommendations.
- Reduced customer satisfaction.
- Incorrect personalization.
- Loss of potential sales in e-commerce systems.

Therefore, accurate WSD directly contributes to search relevance and recommendation accuracy.

#### 4. Suggest an industrial-scale WSD strategy.

An industrial-scale WSD system can use a combination of the following methods:

##### Step 1 – Query preprocessing

Tokenize and normalize the user's search query.

##### Step 2 – Context extraction

Identify surrounding words and domain-specific terms.

##### Step 3 – Knowledge base

Maintain possible meanings and relationships for ambiguous words.

##### Step 4 – Machine-learning model

Use contextual NLP models to estimate the most likely sense.

##### Step 5 – User behaviour analysis

Use clicks, browsing history and previous searches as additional evidence.

##### Step 6 – Feedback mechanism

Continuously update the model using successful and unsuccessful search interactions.

A simplified WSD flow is:

**User Query → Context Extraction → Candidate Senses → Contextual Matching → Correct Sense → Relevant Results**

#### Conclusion

The correct senses are Apple–Technology Brand, Mouse–Computer Device, Java–Programming Language, and Python–Programming Language. The clicked results provide strong contextual evidence for disambiguation. Accurate WSD improves search relevance, recommendations, user satisfaction and e-commerce performance, while incorrect sense selection can produce irrelevant results and reduce system effectiveness.

## Q4. Syntax-Driven Semantic Analysis in Healthcare Information Systems

The healthcare NLP system uses syntactic structures to identify relationships between entities and assign semantic roles. The assessment provides four parsed medical sentences and a set of generated semantic roles.

#### 1. Analyze how syntactic structures contribute to semantic role assignment.

Syntactic analysis identifies the grammatical relationships between words, such as **Subject, Verb and Object**. These relationships provide important information for assigning semantic roles.

Examples:

**Doctor prescribed medicine to patient.**

- Subject → Doctor
- Verb → prescribed
- Object → Medicine
- Recipient → Patient

Therefore:

**Doctor → Agent**

**Medicine → Theme**

**Patient → Recipient**

Another example is:

**Patient reported severe headache.**

- Patient → Subject
- Reported → Verb

- Severe headache → Object/Complement

Semantic interpretation:

**Patient → Experiencer**

**Headache → Symptom**

Thus, syntactic structure provides the foundation for semantic role labeling.

## 2. Determine whether the assigned semantic roles are appropriate.

The given semantic roles are:

| Entity   | Given Role | Evaluation                        |
|----------|------------|-----------------------------------|
| Doctor   | Agent      | Appropriate                       |
| Medicine | Instrument | <b>Needs correction</b>           |
| Patient  | Recipient  | Appropriate in the first sentence |
| Headache | Symptom    | Appropriate                       |

The role **Doctor → Agent** is appropriate because the doctor performs the action of prescribing.

**Patient → Recipient** is also appropriate in:

**Doctor prescribed medicine to patient.**

**Headache → Symptom** is appropriate because headache represents the patient's medical symptom.

However, **Medicine → Instrument** is questionable. In the sentence:

"Doctor prescribed medicine to patient."

Medicine is better represented as the Theme, because it is the entity being prescribed.

Similarly:

**"Medicine reduced blood pressure."**

Medicine is the cause/agent-like entity responsible for the reduction, depending on the semantic-role framework. It should not automatically be classified as an instrument.

Also, the provided role table does not explicitly assign roles to the **Nurse** or the **Patient** in all sentences, even though they participate in the events. The given roles are listed in the assessment.

## 3. Evaluate potential errors from incorrect parsing.

Incorrect syntactic parsing can lead to incorrect semantic interpretation.

For example, if the system incorrectly identifies the subject and object in:

**Doctor prescribed medicine to patient**

it may assign the wrong roles to Doctor, Medicine and Patient.

Possible consequences include:

- Incorrect identification of the healthcare provider.
- Incorrect identification of the patient.
- Incorrect identification of medication.
- Wrong medical event extraction.
- Incorrect clinical information retrieval.
- Errors in medical decision-support systems.

For example, confusing:

**Doctor → Agent**

with:

**Patient → Agent**



could result in an incorrect interpretation of who performed the medical action.

In healthcare applications, such errors can be particularly serious because semantic information may be used for clinical documentation, patient records, decision support and information extraction.

#### 4. Recommend methods to improve syntax-driven semantic analysis.

The following methods can improve healthcare NLP systems:

1. **Medical-domain parsers:** Use parsers trained on medical and clinical text.
2. **Dependency parsing:** Identify grammatical dependencies between medical entities.
3. **Semantic Role Labeling:** Assign roles such as Agent, Patient, Theme, Recipient and Symptom.
4. **Named Entity Recognition:** Identify diseases, medicines, patients, doctors and symptoms.
5. **Contextual language models:** Use models capable of understanding context rather than relying only on individual words.
6. **Medical knowledge bases:** Connect extracted entities with medical concepts and relationships.
7. **Human validation:** Use expert review for high-risk medical interpretations.

A suitable processing pipeline is:

**Medical Report → Tokenization → POS Tagging → Dependency Parsing → Entity Recognition → Semantic Role Labeling → Medical Interpretation**

#### Conclusion

Syntax-driven semantic analysis is important because grammatical structure provides the information required to determine semantic relationships. In the given example, **Doctor as Agent, Patient as Recipient, and Headache as Symptom** are appropriate, while **Medicine as Instrument** should be reconsidered and is better treated as a Theme in the prescription sentence. Improving parsing, semantic role labeling, medical entity recognition and contextual understanding can significantly improve the reliability of healthcare NLP systems

## Rubrics for Calculation

| Criteria                | Weightage | Level 4 (Exemplary)                                                              | Level 3 (Proficient)                                 | Level 2 (Developing)                     | Level 1 (Beginning)                           |
|-------------------------|-----------|----------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------|-----------------------------------------------|
| Understanding of Data   | 25%       | Accurately interprets all data patterns, trends, and relationships               | Interprets data correctly with minor gaps            | Partial understanding; misses key trends | Misinterprets or fails to understand data     |
| Analysis                | 25%       | Provides deep analysis with strong logical reasoning and justification           | Provides reasonable analysis with some justification | Basic analysis with weak reasoning       | No meaningful analysis provided               |
| Application of Concepts | 20%       | Effectively applies relevant concepts/tools to interpret data accurately         | Applies concepts with minor errors                   | Limited application; requires guidance   | Unable to apply concepts to data              |
| Conclusion              | 20%       | Draws clear, meaningful conclusions with strong insights and implications        | Provides valid conclusions with moderate insights    | Conclusions are vague or weak            | No clear or valid conclusions                 |
| Clarity                 | 10%       | Presents interpretation clearly using proper structure, visuals, and terminology | Generally clear with minor issues                    | Lacks clarity or proper structure        | Poor presentation and difficult to understand |

| Q. No. / Criteria Level | Understanding of Data | Analysis | Application of Concepts | Conclusion | Clarity | Sub. Total | Total % |
|-------------------------|-----------------------|----------|-------------------------|------------|---------|------------|---------|
| 1                       |                       |          |                         |            |         |            |         |
| 2                       |                       |          |                         |            |         |            |         |
| 3                       |                       |          |                         |            |         |            |         |
| 4                       |                       |          |                         |            |         |            |         |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
|                   |                    |                  |
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |